

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 13 — Organic Chemistry

Total Marks: 65**Time Allowed: 2h 30m****Version: 1****General Instructions:**

1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Organic compounds are defined as hydrocarbons and their:	isotopes	derivatives	allotropes	isomers only	ⒶⒷⒸⒹ
2	Adjacent members of a homologous series differ by the unit:	–CH ₃	–CH ₂ –	–C ₂ H ₄ –	–OH	ⒶⒷⒸⒹ
3	The general formula of alkanes is:	C _n H _{2n}	C _n H _{2n+2}	C _n H _{2n–2}	C _n H _{2n+1} OH	ⒶⒷⒸⒹ
4	Two compounds with the same molecular formula C ₄ H ₁₀ but different structures are examples of:	allotropy	isomerism	homology	catenation	ⒶⒷⒸⒹ
5	CH ₃ –CH=CH ₂ and CH ₂ =CH–CH ₃ represent which type of isomerism?	chain isomerism	positional isomerism	functional group isomerism	homologous series	ⒶⒷⒸⒹ
6	A compound with functional group –OH attached to an alkyl chain is classified as:	ether	phenol	alcohol	aldehyde	ⒶⒷⒸⒹ
7	The functional group –CHO (carbonyl carbon bonded to one H) identifies:	ketone	ester	carboxylic acid	aldehyde	ⒶⒷⒸⒹ
8	In a ketone, the carbonyl carbon is bonded to:	one H and one alkyl	one –OH and one alkyl	two alkyl groups	one –NH ₂ and one alkyl	ⒶⒷⒸⒹ
9	CH ₃ –O–CH ₃ (dimethyl ether) belongs to which class?	alcohol	ester	ether	ketone	ⒶⒷⒸⒹ
10	The functional group of carboxylic acids is:	–CHO	–CO–	–COOH	–COO–	ⒶⒷⒸⒹ
11	Which compound below is an ester?	CH ₃ COOH	CH ₃ COCH ₃	CH ₃ COOCH ₃	CH ₃ CHO	ⒶⒷⒸⒹ
12	Carbon's ability to bond with other carbon atoms to form long chains is called:	isomerism	catenation	homology	saturation	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define organic chemistry. Why are carbon dioxide, carbon monoxide and carbonates not considered organic compounds?	3	OR	Define organic compounds. Name four elements commonly found in organic compounds besides carbon.	3
2(ii)	Define a homologous series. State four general characteristics of a homologous series.	3	OR	Write the general formulae for alkanes, alkenes, alkynes and alcohols. Using each formula, give the molecular formula of the member with n = 3.	3
2(iii)	What is a structural formula? Write the structural formula and condensed structural formula for propane (C ₃ H ₈).	3	OR	Give the molecular formula, structural formula and condensed formula for butane (C ₄ H ₁₀).	3
2(iv)	Differentiate between saturated and unsaturated hydrocarbons. Give two examples of each.	3	OR	Identify which of the following are saturated and which are unsaturated: (a) CH ₃ –CH ₂ –CH ₃ (b) CH ₃ –C≡CH (c) CH ₃ –CH=CH ₂ (d) CH ₂ =CH–CH=CH ₂ . Give a reason for each.	3
	Define structural isomers. Draw the			Give two alkenes that are structural isomers	

2(v)	structural formulas of the two structural isomers of C ₄ H ₁₀ (n-butane and iso-butane).	3	OR	with molecular formula C ₄ H ₈ . Write their condensed structural formulas and name them.	3
2(vi)	Differentiate between open-chain (acyclic) and closed-chain (cyclic) organic compounds. Give two examples of each.	3	OR	Differentiate between homocyclic (carbocyclic) and heterocyclic compounds. Name one example of each from the textbook.	3
2(vii)	Define a functional group. State three reasons why functional groups are important in organic chemistry.	3	OR	Explain how the functional group present in a molecule determines its chemical and physical properties. Give an example using alcohols and ethers (both have the formula C ₂ H ₆ O).	3
2(viii)	Differentiate between alcohols and ethers: functional group, general formula, and one named example of each.	3	OR	What is a phenol? How does it differ from an alcohol? Classify the following: (a) CH ₃ CH ₂ OCH ₃ (b) C ₆ H ₅ OH (c) CH ₃ OH.	3
2(ix)	Differentiate between aldehydes and ketones. Give the functional group, one example and condensed formula for each.	3	OR	Identify each compound below as an aldehyde, a ketone or a carboxylic acid: (a) CH ₃ COCH ₃ (b) CH ₃ CH ₂ CHO (c) CH ₃ COOH. State the functional group present in each.	3
2(x)	Define carboxylic acids. Write the structural formula of methanoic acid (HCOOH) and ethanoic acid (CH ₃ COOH). Name one daily-life source of each.	3	OR	What are esters? Write the general formula for esters. Give two named examples with their condensed structural formulas.	3
2(xi)	Define amines and amides. State the functional group and general formula for each. Give one named example of each.	3	OR	What is catenation? How does carbon's ability to catenate contribute to the large number of organic compounds? Name two other factors that increase the diversity of organic compounds.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define a homologous series and state its five general characteristics. [3] (b) Using the general formulae, write the molecular formula of: (i) alkane with 5 carbons (ii) alkene with 4 carbons (iii) alcohol with 3 carbons (iv) carboxylic acid with 2 carbons. [2]	5	OR	(a) Define structural isomerism and catenation. Explain with examples how both phenomena contribute to the large number of organic compounds. [3] (b) Draw the structural formulas and write the condensed formulas of all structural isomers of C ₄ H ₁₀ . Name each isomer. [2]	5
3(ii)	(a) Differentiate between saturated and unsaturated hydrocarbons. Describe alkanes, alkenes and alkynes with their general formulae and one example each. [3] (b) Write the structural formula and condensed structural formula of: (i) ethene (ii) ethyne (iii) propene. [2]	5	OR	(a) What are open-chain and cyclic organic compounds? Differentiate between straight-chain and branched-chain compounds. Give the condensed structural formula for iso-butane and neo-pentane. [3] (b) Identify each structure below as straight-chain, branched-chain or cyclic: (i) CH ₃ –CH ₂ –CH ₂ –CH ₃ (ii) CH ₃ –CH(CH ₃)–CH ₃ (iii) cyclopropane (iv) benzene. [2]	5
3(iii)	Identify the functional group present in each of the following compounds, name the class it belongs to, and write the general formula for that class: (a) CH ₃ OH (b) CH ₃ –O–CH ₃ (c) CH ₃ CHO (d) CH ₃ COCH ₃ (e) CH ₃ COOH. [5]	5	OR	(a) Define a functional group. State three reasons why functional groups are important. [2] (b) Identify the class of each compound and its functional group: (i) CH ₃ COOCH ₃ (ii) CH ₃ NH ₂ (iii) CH ₃ CONH ₂ (iv) C ₆ H ₅ OH (v) CH ₃ CH ₂ I. [3]	5
3(iv)	(a) Describe alcohols, ethers, carboxylic acids and esters: give the functional group, general formula and one named example for each. [4] (b) Ethanol (CH ₃ CH ₂ OH) and dimethyl ether (CH ₃ –O–CH ₃) have the same molecular formula C ₂ H ₆ O. Are they isomers? Explain which class each belongs to. [1]	5	OR	(a) Describe amines and amides: functional group, general formula and one named example for each. [2] (b) Classify each compound as amine, amide, ester or carboxylic acid, and write the functional group: (i) CH ₃ NH ₂ (ii) CH ₃ CONH ₂ (iii) CH ₃ COOC ₂ H ₅ (iv) CH ₃ CH ₂ COOH. [3]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 13 — Organic Chemistry (pp. 174-190)

— End of Question Paper —